Clinical background

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0.1 Clinical background and data sources

Our analysis combines two data sources on the treatment of HIV-1 infection. The randomized component is the AIDS Clinical Trials Group Study 175 (ACTG 175), a double-blind trial that enrolled 2,139 HIV-1–infected adults with CD4 cell counts between 200 and 500 cells/mm^3 and randomized them to one of four nucleoside reverse-transcriptase inhibitor regimens: zidovudine (ZDV) monotherapy, didanosine (ddI) monotherapy, ZDV combined with ddI, or ZDV combined with zalcitabine (ddC) (Hammer et al., 1996). The observational component is drawn from the Multicenter AIDS Cohort Study (MACS), a long-running prospective cohort of men who have sex with men that has documented the natural and treated history of HIV-1 infection since 1984 (Kaslow et al., 1987). We restrict both sources to a comparable pre-HAART calendar window and to participants meeting the trial's eligibility criteria, and we define a binary treatment contrast: ZDV monotherapy versus the pooled alternative of ddI monotherapy, ZDV+ddI, or ZDV+ddC. The outcome is event-free survival, defined as the time to the first of a $\geq 50\%$ decline in CD4 count, an AIDS-defining event, or death.

The clinical evidence on this treatment contrast is well established. ACTG 175 found that ZDV monotherapy was inferior to each of the other three regimens, which were in turn comparable to one another: ddI monotherapy, ZDV+ddI, and ZDV+ddC each delayed disease progression and prolonged survival relative to ZDV alone (Hammer et al., 1996). The Delta trial reached the same conclusion, showing that combinations of ZDV with ddI or ddC were superior to ZDV monotherapy (Delta Coordinating Committee, 1996). The pharmacological explanation is the rapid emergence of resistance under ZDV monotherapy: HIV-1 reliably accumulates thymidine-analogue mutations in reverse transcriptase within months of treatment, eroding any initial benefit (Larder et al., 1989). While zidovudine had earlier been shown to confer a short-lived advantage over deferred treatment in symptom-free individuals (Concorde Coordinating Committee, 1994), this transient benefit reflects a comparison against no treatment and does not extend to a comparison against ddI- or combination-based regimens. Pooling the three superior arms into a single comparator is therefore both clinically and statistically justified.

This contrast is well suited to studying the integration of randomized and observational evidence precisely because the two sources are expected to disagree. In observational practice, the choice of regimen was driven by disease severity, immunologic trajectory, and prior treatment experience, so that patients with poorer prognosis were preferentially escalated to ddI-containing or combination therapy. This channeling induces confounding by indication that operates opposite to the true treatment effect, and can produce a spurious early survival advantage for ZDV monotherapy in the observational cohort—together with an apparent attenuation of that advantage over follow-up, as the mis-sorted high-risk patients experience events and the susceptible pool is depleted (Hernán, 2010). Neither feature is present under randomization. The discrepancy between the unconfounded trial estimate and the confounded observational estimate is precisely the quantity that the confounding function in our framework is designed to capture.

- ACTG 175 was a double-blind randomized trial of 2,139 HIV-1–infected adults with CD4 counts of 200–500 cells/mm^3, randomized to ZDV monotherapy, ddI monotherapy, ZDV+ddI, or ZDV+ddC; the primary endpoint was event-free survival, i.e. time to a $\geq 50\%$ CD4 decline, an AIDS-defining event, or death (Hammer et al., 1996).
- The trial's principal finding was that ZDV monotherapy was inferior to each of the other three regimens, which were statistically comparable to one another — providing the clinical justification for collapsing them into a single comparator arm (Hammer et al., 1996).
- The Delta trial independently corroborated this result, showing ZDV+ddI and ZDV+ddC superior to ZDV monotherapy (Delta Coordinating Committee, 1996).
- The inferiority of ZDV monotherapy is mechanistically explained by the rapid emergence of thymidine-analogue resistance mutations in HIV-1 reverse transcriptase, which erode the initial treatment response within months of starting therapy (Larder et al., 1989).
- The early, short-lived benefit of ZDV documented in Concorde was a comparison against *deferred or no* treatment in symptom-free individuals; it does not imply any advantage of ZDV monotherapy over ddI- or combination-based regimens, and so does not apply to the present treatment contrast (Concorde Coordinating Committee, 1994).
- MACS is a prospective observational cohort of men who have sex with men that has followed the natural and treated history of HIV-1 infection since 1984; regimen choice was made in routine clinical care rather than randomized (Kaslow et al., 1987).
- In observational practice, regimen choice tracked disease severity, CD4 trajectory, and prior treatment experience, so that poorer-prognosis patients were preferentially escalated to ddI-containing or combination therapy (confounding by indication).
- This channeling sorts high-risk patients into the comparator arm and can make ZDV monotherapy appear spuriously favorable in the observational data — opposite in direction to the randomized result — with an apparent attenuation of that advantage over follow-up as susceptible patients are depleted (Hernán, 2010).
- The introduction of protease inhibitors and HAART in 1996 redefined standard of care; both data sources must be restricted to a comparable pre-HAART calendar window for the treatment definitions to remain meaningful and exchangeable.
- ACTG 175 was stratified by prior antiretroviral use; ZDV-experienced patients are more likely to harbor pre-existing resistance, making prior ARV exposure a clinically motivated candidate effect modifier for the treatment effect.
- In the observational cohort, regimens were switched as disease progressed; a baseline-fixed treatment definition therefore leaves later follow-up subject to switching-induced selection.
- Regimen toxicity profiles differ (ZDV: bone-marrow suppression; ddI: pancreatitis and peripheral neuropathy; ddC: peripheral neuropathy), which can contribute to early, regimen-specific attrition.

0.2 The error distribution

Two features of the application dictate the form of the error distribution: it must be nonparametric, and it must be specific to the data source.

Why nonparametric. The event-free survival endpoint is a composite, defined as the first occurrence of a $\geq 50\%$ decline in CD4 count, an AIDS-defining event, or death (Hammer et al., 1996). The time to the first of several biologically distinct events does not follow any standard parametric family, and the log-normal, Weibull, and log-logistic errors implied by the usual parametric accelerated failure time models cannot be expected to capture its shape. Crucially, in this setting error misspecification is not innocuous. A parametric accelerated failure time model fully specifies the likelihood, so an incorrect error shape is accommodated by distorting the location parameters—the baseline function m_0, the treatment effect τ, and the confounding function c. Because these location parameters are precisely the causal estimands of interest, a nonparametric error is required to protect them from shape misspecification, rather than merely to improve the fit of the survival curve.

Why source-specific. A single nonparametric error, shared between the trial and the cohort, still imposes the assumption that the two samples have an identical residual distribution. This homogeneity assumption is difficult to defend here. First, the decomposition is a working model marginal over the unmeasured confounders U; because treatment is selected on U in the observational study but randomized in the trial, the residual distribution of the treated observational arm is reshaped relative to the trial, and this reshaping affects higher moments that the confounding function $c(X)$, capturing only the location, does not absorb. Second, the endpoint is ascertained differently—on a protocol schedule with adjudication in the trial, and by reconstruction from approximately six-monthly cohort visits in the observational study, which coarsens event times. Third, the censoring and follow-up regimes differ, the cohort being additionally subject to loss to follow-up and to truncation at the onset of the HAART era. Fourth, the observational population is broader and less selected for adherence than the trial population. These mechanisms predict differences in the *shape* of the residual distribution—its skewness, tails, and modality—and not merely its scale, so that a shared error with a source-specific variance would also be inadequate. At the same time, the two samples concern the same disease, endpoint, and time scale, so their error distributions are related rather than arbitrary. The hierarchical Dirichlet process is the appropriate model for this situation: it equips each source with its own error distribution while sharing a common pool of mixture components, and it reduces to a single shared distribution when the two samples are in fact homogeneous. The hierarchical specification therefore relaxes the homogeneity assumption without requiring that the sources be shown to differ in advance.

The error distribution: the argument.

Part A — the error must be nonparametric.

- **R1.** The endpoint is a composite—the first of a $\geq 50\%$ CD4 decline, an AIDS-defining event, or death—and the time to the first of such heterogeneous events has no standard parametric form.

- **R2.** In an AFT model the likelihood is fully specified, so a misspecified error is absorbed by the location parameters m_0, τ, c. Since these are the causal estimands, parametric misspecification biases the targets of inference.

Part B — the error must be source-specific (HDP, not a single DP). A single DP assumes the trial and the cohort share one error distribution; this homogeneity assumption fails for four reasons.

- **R3 (structural).** The model is marginal over the unmeasured U; treatment is selected on U in the cohort but randomized in the trial, reshaping the cohort's residual distribution. $c(X)$ corrects the location of this distortion, not its shape.
- **R4 (measurement).** The endpoint is adjudicated on a protocol schedule in the trial but reconstructed from six-monthly visits in the cohort, coarsening event times.
- **R5 (design).** Censoring differs: administrative in the trial; loss to follow-up and HAART-era truncation in the cohort.
- **R6 (population).** The cohort is broader and less adherence-selected than the trial.

Two further steps complete the argument.

- **R7.** R3–R6 alter the *shape* of the error—skew, tails, modality—not merely its scale; a shared DP with a source-specific variance is therefore also insufficient.
- **R8.** The two sources share disease, endpoint, and time scale, so independent DPs would discard common structure.

Conclusion. R1–R2 require a nonparametric error; R3–R8 require it to be source-specific with partial pooling. The hierarchical Dirichlet process is the implied model—source-specific error distributions over a shared atom pool—and it reduces to a single DP when homogeneity holds.

Important questions.

- From when do we start counting time? For the RCT since randomisation; how about the RWD?
- Go into the selection of the RCT—restrict the RWD to a similar cohort?
- How can we use the WIHS data? Merge it with the MACS, treat it as a separate source, or use it as a validation set?
- Interval censored: currently the left interval point is registered as the event time.

References

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